

Summary

Rhett D Harrison

2026-06-19

Headline conclusions

- **CFM is broadly working, but governance foundations are fragile.** Clear majorities of communities report reduced deforestation, lower fire frequency and recovering wildlife populations under CFM. These gains are real, but they rest on institutional foundations that are incomplete in many CFMGs, and are currently held together primarily by conservation NGOs and carbon enterprises rather than by the state.
- **Satellite evidence confirms fire and biomass gains.** Remote sensing analysis independently corroborates community perceptions: fire incidence, burned area and fire radiative power all declined significantly following CFMG establishment, and above-ground biomass increased in both CF cores and surrounding buffer zones. There is no evidence of displacement of burning into buffer areas.
- **Deforestation rates are low but not declining.** Tree cover loss rates within CFs were already very low before establishment and did not decrease significantly afterwards. There is a concern that deforestation may have increased in the 1 km buffer zones, though this result requires longer monitoring to confirm.
- **Governance processes are incomplete across most CFMGs.** Many CFMGs lack bank accounts, receipt books, annual workplans and business plans. Most are overdue for elections and thus non-compliant with their own bylaws. Without sound governance, CFMGs cannot sustainably manage finances, account for income or maintain compliance with forest management licenses.
- **State oversight is insufficient.** The Forestry Department (FD) has a statutory duty to support and monitor CFMGs, but District Forestry Officer (DFO) visits are infrequent across most of the country. Conservation NGOs and carbon enterprises partly fill this gap, but coverage is uneven and a substantial number of CFMGs receive no meaningful external support.

- **Carbon income is transformative but narrowly distributed.** Approximately one third of CFMGs had received carbon payments at the time of the survey. Employment has grown under CFM but is concentrated in low-paying roles, is male-dominated, and is absent in many communities.
- **The economic potential of CFM is massively under-realised.** Beyond carbon and honey, the considerable wealth of Zambia’s forest products is almost entirely unexploited as a basis for community enterprise. A far more diversified economic model is achievable and necessary for long-term viability.
- **Equity outcomes are broadly positive, but warning signals exist.** Most female FGD groups report equitable benefit sharing and meaningful participation in decision making — a more positive picture than expected. However, youth groups are consistently more critical, and a minority of communities show signs of elite capture, lack of transparency and exclusion that must be addressed.
- **Communities are clear-eyed about both gains and gaps.** Community feedback on what is working matches the quantitative evidence. Their identification of what needs improving provides a practical agenda for action.

Survey details

Community forest management (CFM) has expanded rapidly in Zambia since receiving enabling regulations under the Forest Act in 2018. As of 2025, 351 Community Forest Management Groups (CFMGs) preside over approximately 9.4 million hectares of community forests across all ten provinces. In contrast to many other countries, CFM in Zambia involves the lawful transfer of exclusive economic rights over forest resources to CFMGs — a provision that has made Zambia an attractive destination for voluntary carbon markets and conservation investment, and has driven the rapid expansion of the programme.

This report presents findings from a survey of 80 CFMGs conducted across eight provinces in 2024. The two provinces omitted — Luapula and Copperbelt — had very few CFMGs at the time of the survey and their exclusion does not affect national representativeness. Five CFMGs included in the original sample were found to be entirely non-functional and were excluded, leaving a final analytical sample of 75 CFMGs. The sample spans a wide range of CF sizes (from approximately 20 ha to over 80,000 ha) and ages (from one to approximately seven years since establishment; [age distribution](#)), and reflects the national age structure of the programme, in which most CFs are fewer than three years old.

The survey employed a multi-perspective focus group discussion (FGD) design. Four structured FGDs were conducted at each CF: with the CFMG executive committee, community

men, community women and community youth ([survey coverage](#)). This design enables the systematic comparison of perspectives across stakeholder groups and is the methodological feature most critical to interpreting the equity and inclusion findings. Near-complete coverage of all four FGD types across the 75 CFs means that differences in responses among groups can be interpreted with confidence. Individual interviews were conducted with CFMG executive members, DFOs, traditional leaders, forest user-group chairs and external partners (conservation NGOs and carbon enterprise representatives), providing an additional layer of evidence from different vantage points.

The geographic spread of the sample — covering Northwestern, Western, Luapula (substituted), Northern, Muchinga, Eastern, Southern and Central provinces ([coverage by province](#)) — ensures that findings reflect diverse ecological, economic and socio-political contexts. The gender breakdown of interviewees is male-biased, as expected given that men dominate most CFMG stakeholder roles; however, the dedicated female FGD in each CF ensures that women’s perspectives are systematically captured. Interviewees’ years of involvement in CFMGs are broadly consistent with the age of the programme, and their time living in their communities confirms they are well-embedded members rather than recent arrivals.

In sum, the sample provides a robust, nationally representative foundation for drawing inferences about CFM implementation in Zambia. The multi-perspective design is particularly well suited to examining the equity and governance dimensions of CFM, which are among the most important — and most contested — aspects of the programme.

Governance

The governance of CFMGs is assessed across five domains: legal processes, governance processes (elections, meetings, financial management), business processes (workplans, budgets, business plans), external support, and community buy-in. The most consistent pattern across all three process domains is that foundational steps have been taken by most CFMGs, but the more demanding ongoing requirements are frequently incomplete.

Legal processes are the most advanced domain. The large majority of CFMGs in the sample have obtained a recognition letter and signed a forest management agreement — the foundational steps for establishing a CF ([legal process completion](#)). However, later and more administratively demanding processes, including the registration certificate, forest-use permit and benefit-sharing agreement, are absent in a substantial proportion. These are not minor procedural details: without a registration certificate and forest-use permit, a CFMG’s legal basis for managing its forest and excluding external actors is weakened, and its ability to negotiate with carbon buyers and other commercial partners is compromised.

Governance processes are more unevenly implemented. Most CFMGs have held at least one election and have term limits for executive officers ([election recency](#)), and most hold CFMG

meetings with reasonable frequency. However, several basic management actions are missing in a large proportion of CFMGs: bank accounts have not been opened, receipt books are not maintained, and annual financial and technical reports are rarely produced ([governance process rates](#)). These are not peripheral concerns. Without bank accounts, carbon payments and other income cannot be transparently managed. Without receipt books and financial reporting, there is no basis for demonstrating accountability to community members or to external partners. The fact that most CFMGs are overdue for elections is a particular concern. Timely elections are a basic requirement of the CFMG bylaws and constitution; overdue elections undermine democratic legitimacy and, in some cases, appear to reflect deliberate retention of control in anticipation of carbon income.

Business processes are the weakest domain of all. Annual workplans, budgets and business plans are present in only a minority of CFMGs ([business process rates](#)). Without these planning and financial management tools, CFMGs are poorly positioned to manage resources sustainably, account for income or sustain compliance with their forest management license.

External support presents a mixed picture. Most CFMGs are rarely or never visited by DFOs, despite the statutory duty of the FD to implement the CFM policy ([DFO visit frequency](#)). This gap is partly compensated by conservation NGOs and carbon trading enterprises, many of which visit their partner CFMGs regularly. However, this support is not universal: a significant proportion of CFMGs in the sample had no external partner of any kind ([external partner coverage](#)). Where NGO or carbon enterprise support is present, it has funded real improvements in forest management — patrol systems, fire management and agricultural extension. The current model, in which CFM is effectively co-governed by the community and the private sector rather than the state, may be adequate where active and capable partners are present, but leaves many CFMGs without meaningful support.

Interest in expanding the community forest, which is used as an indicator of community satisfaction with CFM, is highest among CFMG executives and lowest among community youth ([interest in expansion by stakeholder group](#)). Among individual interviewees, NGO representatives are significantly more enthusiastic about expansion than traditional leaders or community members. This divergence — consistent with the equity findings discussed below — suggests that CFMG leadership and external partners have a more optimistic assessment of CFM outcomes than the broader community. If not addressed, this gap risks eroding community legitimacy and long-term sustainability of CFM.

In summary, CFMG governance is still maturing. The legal framework for CFM is largely in place, but institutional and business management processes lag behind. Strengthening financial management, ensuring timely elections, expanding DFO engagement and closing the gap between executive and community perspectives will be essential if CFM is to fulfil its potential.

Sustainable Forest Management

Across all SFM indicators, the evidence points to meaningful improvements under CFM. Clear majorities of community FGD groups report that deforestation has decreased ([reduced deforestation](#)), charcoal production has declined, fire frequency has fallen ([reduced fire](#)) and wildlife populations have increased ([wildlife recovery](#)) within their community forest. These are substantial changes in a programme that, for most communities, has been operating for fewer than five years.

Critically, the large majority of FGD groups also report that deforestation and charcoal production are decreasing in areas *outside* their community forest ([deforestation outside CF](#)). This is an important result: it indicates that CFM is not simply displacing pressure from protected areas into adjacent landscapes, but is generating broader environmental benefits in the surrounding region. The absence of leakage is consistent with the remote sensing findings described in the following section.

Community female FGD groups tend to report particularly positive changes in SFM outcomes — a pattern that recurs across multiple indicators and domains. This may reflect that women, who are traditionally highly dependent on forest resources for household provisioning but have historically had limited say in management decisions, have gained meaningful voice through CFM’s formal governance structures. Following up this finding to understand what is driving women’s more positive perception of SFM outcomes would be valuable.

Among individual interviewees, DFOs and NGO representatives tend to report somewhat more conservative assessments of positive change than community members. This may reflect a broader professional perspective on forest management, or it may reflect that professionals are less aware of the day-to-day improvements visible to those living within and adjacent to the community forest. Longer-term monitoring will be required to disentangle these possibilities.

Restoration activities — natural regeneration protection and, in a few cases, planting — are occurring in some CFMGs, but are not widespread ([restoration uptake](#)). Given the Zambian government’s restoration commitments and the central role that CFM could play in delivering them, this is a significant missed opportunity. A dedicated restoration component — covering natural regeneration, indigenous planting and, where appropriate, exotic species — should be integrated into forest management plans for all CFMGs. Similarly, agroforestry and plantation establishment, which could both reduce pressure on natural forests and generate valuable additional income streams, are practiced by only a small number of CFMGs.

In summary, CFM in Zambia is driving real improvements in sustainable forest management across the large majority of surveyed communities. However, these improvements are not ubiquitous, and a minority of CFMGs are not making progress. Identifying underperforming CFMGs, diagnosing the causes and implementing corrective action will be essential. Equally, integrating restoration, plantation and agroforestry components into the CFM model would substantially strengthen its contribution to national SFM targets.

Remote Sensing analysis

The remote sensing analysis provides independent, satellite-based evidence on four key indicators of CFM effectiveness: fire incidence, burned area, fire radiative power and above-ground biomass (AGB). In addition, tree cover loss data from the Global Forest Canopy dataset (Hansen et al.) are used to assess deforestation rates within CF cores and the surrounding 1 km buffer zone. The analysis uses a before-after, control-impact design that compares outcomes within CF cores before and after establishment, with buffer zones as an approximate control.

Across all three fire metrics — fire detection density ([Figure](#)), burned area fraction ([Figure](#)) and fire radiative power ([Figure](#)) — there were statistically significant reductions within CF cores following establishment. These results constitute robust, multi-indicator satellite confirmation that CFM has led to meaningful improvements in fire management at a landscape scale. Importantly, none of the fire models showed a significant After $\times$ Buffer interaction, indicating that there was no measurable displacement of burning into surrounding buffer zones. This absence of leakage from CF cores to adjacent areas is consistent with community perceptions documented in the SFM chapter, where a clear majority of FGD groups reported that fire frequency was also declining outside their CF.

According to the Hansen Global Forest Canopy dataset, tree cover loss rates within CF cores did not decrease significantly following establishment ([tree cover loss](#)). This result may initially appear contradictory given the strongly positive community perceptions, but several explanations are plausible. First, tree cover data are often unreliable at small spatial scales in miombo woodlands, where canopy cover is naturally sparse and can vary seasonally due to the relative strength of the grass versus tree canopy signal. Second, baseline deforestation rates within CF cores were already very low (median of approximately 0.14% per year) before establishment, leaving limited scope for measurable further decline. The directional trend is in the right direction, even if the change is statistically modest. Meanwhile, communities may be perceiving real improvements through indirect indicators — such as reduced charcoal production, fewer incursions or recovery of the forest understorey — that are not captured by the canopy loss metric. The significant increase in tree cover loss in the 1 km buffer zones (from a median of 0.26% before CF establishment to 0.49% afterwards) is a concern, but should be interpreted cautiously given the measurement reliability issues noted above. Continued monitoring over a longer period is recommended before strong conclusions are drawn.

Above-ground biomass increased significantly in both CF cores and buffer zones following CFMG establishment ([above-ground biomass](#)). For dry seasonal forests such as the miombo woodlands that dominate Zambia's community forests, biomass is likely a more sensitive and reliable indicator of ecological condition than canopy cover loss. Biomass monitoring can detect incremental changes caused by shifts in tree recruitment and growth, which are not captured

by gross deforestation metrics. The observed biomass gains are consistent with — and likely substantially attributable to — the documented reductions in fire disturbance. Fire both kills trees directly and suppresses woody recruitment; even modest reductions in annual burning can produce measurable biomass accumulation over a few years.

In summary, the remote sensing evidence presents a nuanced but largely positive picture of CFM effectiveness. CFM has demonstrably reduced fire disturbance and, as a consequence, has produced measurable gains in above-ground biomass across both CF cores and their immediate surroundings, without evidence of displacement of burning into adjacent areas. The deforestation signal is more ambiguous and warrants continued monitoring, but it does not negate the positive evidence from fire and biomass indicators. The economic drivers of deforestation — charcoal production, agricultural clearance — remain largely unaddressed by current CFM practices, and this is the domain where further policy attention is most needed.

Economic transformation

The economic outcomes of CFM are real but narrowly distributed, and the model's full potential remains far from being realised. The analysis covers five domains: access to forest resources, employment, carbon income, non-timber forest products and other income-generating activities, and opportunity costs.

Most FGD groups and individual interviewees indicate that CFM has not increased access to natural resources for either household use or sale. This is perhaps unsurprising: communities were already harvesting forest products before CFM was established, and formalisation does not necessarily change the volume or diversity of what is harvested. However, approximately one third of respondent groups felt that access had improved, likely reflecting the formalisation of community rights and the greater security of tenure that comes with a registered community forest.

Employment has grown under CFM ([employment by FGD group](#)), but the picture is more complicated than the headline numbers suggest. The most significant employment gains are in the roles of Honorary Forest Officers (HFOs) and community scouts — positions that are low-paying, predominantly male and absent or minimal in many communities. Community female FGD groups were the most likely to report no change in formal employment, and community youth groups were the most likely to report gains — a difference that likely reflects the demographics of the HFO and patrol roles that CFM has generated. While new employment is a genuine achievement, the scale and quality of that employment falls well short of what a thriving forest-based economy should deliver.

Carbon income is the centrepiece of the CFM economic model in Zambia, and its impact is significant where it has materialised. Approximately one third of CFMGs in the sample had

received carbon payments at the time of the survey ([carbon finance by province](#)). The activities funded by carbon trading enterprises — patrol systems, fire management, climate-smart agriculture extension — represent a real injection of capital into forest margin communities. However, the fact that a substantial number of community youth FGD groups were unaware of the carbon agreement ([carbon awareness by FGD group](#)) is concerning: it suggests that carbon enterprises may not have conducted a fully inclusive FPIC process, and that CFMG executives may not be communicating adequately with their communities. This information asymmetry is a governance risk that could erode community support for CFM over time.

Beyond carbon and honey, the economic potential of Zambia’s community forests is almost entirely unrealised. A very wide variety of forest products are harvested from community forests — including timber, wild fruits, medicinal plants, mushrooms, caterpillars, bamboo, resins and thatching grass — but overwhelmingly for household use or local sale with minimal value addition. Only a small number of CFMGs are partnering with any kind of enterprise ([enterprise collaboration](#)), and where they are, it is almost exclusively for honey. The communities themselves identify a rich diversity of products — prominently including timber and wild fruits as well as mushrooms and honey — that could be developed into community enterprises ([unrealised opportunities](#)). The potential to layer multiple income streams on top of one another, supply urban and regional markets with value-added products, and create more diverse and stable income for community members is enormous. CFM has the legal and institutional scaffolding to support this kind of diversified forest economy; what is lacking is the investment, technical support and market linkages needed to activate it.

Opportunity costs of CFM — foregone income, restricted access to previously used areas, time and resource costs of participation — are recognised by communities ([opportunity costs by FGD group](#)), and are important to acknowledge. The distribution of these costs is uneven: opportunity costs fall most heavily on households that were most dependent on forest clearing or other activities that are now restricted, and these households are disproportionately among the most economically vulnerable. Properly accounting for and mitigating opportunity costs — including through explicit provision in benefit-sharing arrangements and business plans — should be a standard part of CFMG management.

In summary, CFM has achieved real but limited economic transformation. The economic model currently relies too heavily on carbon and a single NTFP (honey), leaving most communities with a narrow income base that is vulnerable to market fluctuations and partner decisions. A far more diversified economic model — one that exploits the full range of forest products, develops community enterprises, and actively links communities to urban, regional and international markets — is both achievable and necessary.

Equity and Inclusion

The equity and inclusion chapter investigates a wide range of outcomes: benefit sharing, opportunity costs, gender equity, transparency, grievance and redress, and poverty impacts. The overarching finding is that equity outcomes under CFM are broadly positive, but the picture is uneven and contains important warning signals.

The most striking pattern across all equity indicators is a systematic divergence in perspective by FGD group. Community female and youth groups consistently report more negative assessments of equity outcomes than CFMG executive and community male groups — across virtually every indicator, including equitable benefit sharing (Figure), women benefiting from CFM (Figure), women’s participation in decision making (Figure), and transparency in benefit sharing (Figure). This divergence could reflect information asymmetry: CFMG executives, who are better informed about formal governance arrangements, may genuinely perceive a more equitable distribution than is visible to other community members. Alternatively — and perhaps more plausibly in some communities — it may reflect that the benefits and costs of CFM are in fact being distributed unevenly, in ways that are visible to those outside the executive circle but not acknowledged by those within it. Elite capture and corruption cannot be ruled out in a subset of cases.

Notwithstanding these concerns, the overall picture is more positive than might have been expected. A clear majority of all FGD groups — including community female groups — report that benefits are being shared equitably and that women benefit from CFM. Most female groups state that women do not disproportionately bear opportunity costs, and that women participate in CFM decision making. This is a genuinely surprising and encouraging finding in a country where natural resource management has traditionally been the purview of male community members. The formal governance structures mandated by CFM — including elections, executive positions with required female representation, and documented benefit-sharing arrangements — appear to have given women a degree of recognition and voice that they did not previously have. CFM has, in this respect, served as a partial tool of institutional gender mainstreaming in forest management.

Community youth groups present a consistently more critical perspective across all domains. Youth are the most sceptical about CFM’s benefits, the most concerned about elite capture and transparency, and the most likely to report that CFM has not delivered meaningful livelihood improvements for their demographic. These concerns are consistent with the economic transformation findings: CFM employment is concentrated in roles (HFOs, community scouts) that are predominantly occupied by older community members, and opportunities for youth-specific enterprise development are limited. Youth dissatisfaction, if unaddressed, poses a genuine long-term risk to community support for CFM.

Carbon benefit-sharing agreements are structured almost entirely around revenue sharing with little provision for employment, infrastructure investment or targeted livelihood support.

While revenue sharing delivers a direct financial return when payments materialise, communities that have not yet received payments — which is the majority — have little tangible benefit to show. Diversifying the components of company-community agreements to include employment targets, skills development, and livelihood support would broaden both the distribution of benefits and community buy-in across diverse stakeholder groups.

The poverty impacts of CFM — primarily through increased household incomes and new income-generating activities ([poverty impact](#)) — are broadly consistent with the economic transformation findings. They are real but limited, and the barriers that prevent the most vulnerable households from benefiting — limited access to resources, capital and markets — are structural and will require deliberate policy attention to overcome.

In summary, CFM has made meaningful progress on equity and inclusion, particularly with respect to women’s voice in forest management. However, the divergence between the perspectives of community female and youth groups on one side and CFMG executives and males on the other is the most important equity signal from this survey. Strengthening transparency, grievance and redress mechanisms, ensuring that youth groups are meaningfully engaged in benefit sharing, and monitoring for elite capture are essential to ensuring CFM delivers equitable outcomes across the full diversity of community stakeholders.